

MS

Chem 4

4

(a) (i) Sewage

Is a variable and complex mixture of organic matter (fats, soap, urine, oil, food material and faeces) and inorganic matter (nitrates, phosphates, clay, sand and a variety of metal ions) with water being greater than 99% AW; [1]

BOD

Is a measure of the amount of oxygen in ppm which is required by bacteria to metabolise the organic matter over a 5-day period/a measure of organic matter present in water as determined by amount of oxygen taken up by the sample / AW; [1]

- (b) (i) Sedimentation, screening of floating materials; Al^{3+} as flocculation agent (forms ppt of $Al(OH)_3$ which absorbs finely suspended matter before sinking; [1]
- (ii) 1. to remove dissolved organic matter / reduce BOD; [1]
2. air passed into aeration tank encourages growth of aerobic bacteria; [1]

Which oxidises dissolved organic matter to CO_2 and H_2O ; [1]

- (iii) - ~~Traces of organic matter~~ are removed by addition of charcoal; [1]
- Phosphates are removed by precipitation using Ca^{3+} lime, alum or $FeCl_3$; [1]
- Nitrates are removed by addition of lime to convert nitrates into $NH_{3(g)}$ which is discharged into the atmosphere / Denitrifying bacteria can be used to form volatile $N_{2(g)}$; Any two [2]

- (c) Cancer development due to formation of chloroalkanes; [1]
- [Total: 10]

- 5 (a) (i) A gas in the atmosphere which absorbs and emits radiation causing global warming/AW; [1]
- (ii) Water vapour, methane, dinitrogen oxide, CFC(s) and ozone: (any two) [2]
- (iii) Water vapour → physical change of state/condensation/precipitation;

Methane → chemical reaction / oxidation by reaction with hydroxyl free radicals ($OH\cdot$) and degradation to CO_2 ;

Carbon dioxide → photosynthesis / CO_2 dissolving in oceans to form carbonic acid (or HCO_3^- / CO_3^{2-})

N-oxides/ SO_2 → absorbed by rain water

Ozone: change to O_2 ; Any two [2]

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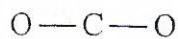
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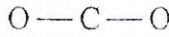
- (b) Burning of fossils fuels / cement manufacture leads to higher carbon dioxide concentration; *1 point of time* [1]

- (c) CO₂ traps energy and the molecule vibrates with a frequency within the infra-red region of the electromagnetic spectrum/AW; [1]

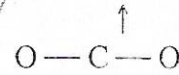
Vibration modes are



assymetrical stretch



symmetrical stretch



bending mode *any one* [1]

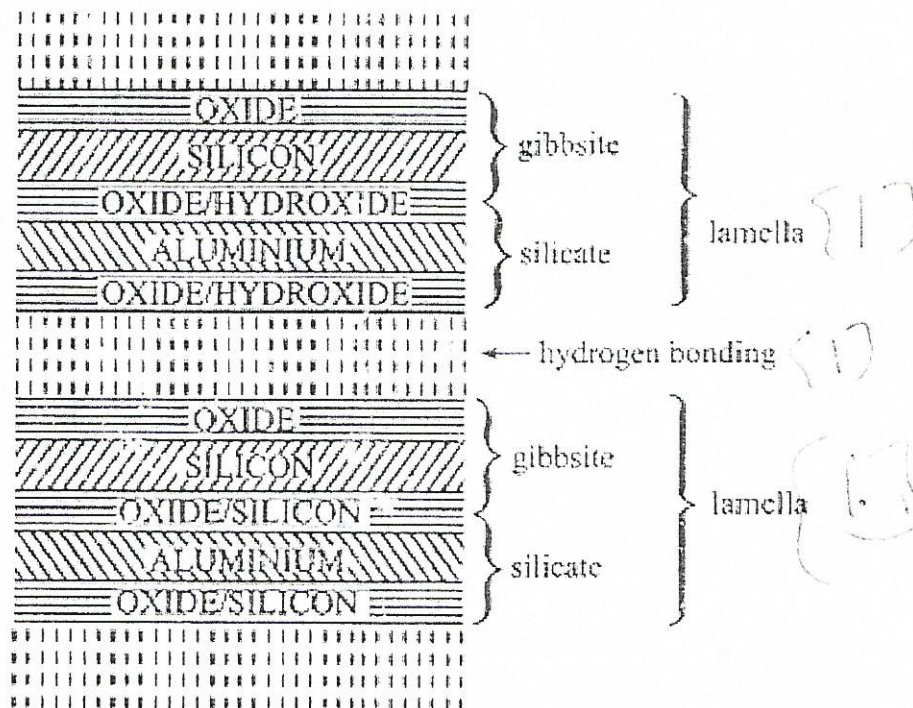
Only the assymetrical stretch and bending mode involve change in dipole enabling absorption of energy in infra-red region/AW; [1]

Causing an increase in temperature of the troposphere; [1]

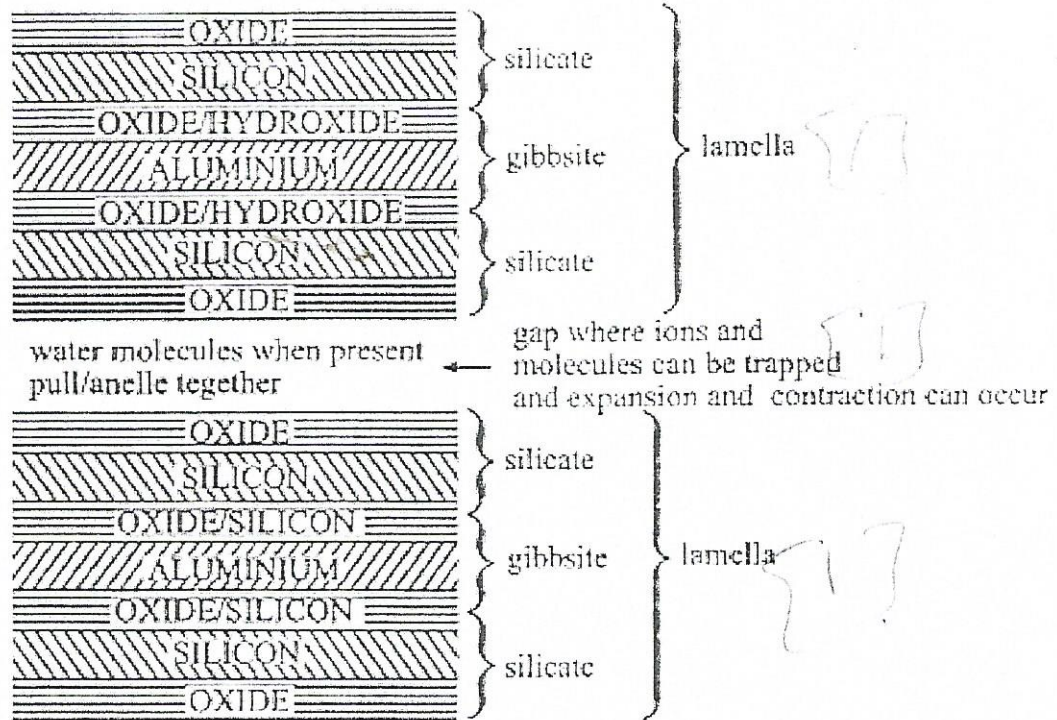
[Total: 10]

6

(a)



1:1 clay



2:1 clay

- (b)
- NH_4^+ and K^+ are held by clay cation exchange sites (CES); [1]
 - these ions displace the ions already present; [1]
 - ~~the~~ NH_4^+ and K^+ remain available rather than being washed away; [1]
 - formation of acid rain produces H^+ which can be held at CES in high concentrations, thereby displacing NH_4^+ and K^+ ; [1]
- [Total: 10]

- 7 (a)
- (i) Eutectic point / composition of liquid mixture with lowest freezing point; [1]
- (ii) Eutectic composition / 62% Sn and 38% Pb ; (A) *wide fraction* [1]
0.62 0.38
- (iii)
- Eutectic temperature / 183°C ; [1]
 - Because eutectic mixture has lowest melting point; [1]
- (iv)
- Freezing point of Pb is lowered; [1]
 - Interposition of Sn atoms between Pb atoms lowers the v.p. of Pb / increases distance between Pb atoms $\therefore$ lower temperature is required to bring them close enough for solidification to occur; [1]
- (b)
- (i) T_2 / 1260°C ; *1250* [1]
- (ii)
1. *56%* 58% Ni / 42% Cu ; *56% 44%* [1]
 2. 30% Ni / 70% Cu (or both); *33% Ni 67%* [1]

- (c) - Unlike the phase diagram for Pb-Sn, phase diagram for Cu-Ni system has no eutectic point /
- For Pb-Sn system addition of tin lowers F.p. of Pb until eutectic point is reached whereas addition of Ni to Cu increases the f.p. of Cu/AW; Any 1) [1]
[Total: 10]

8 (a) Partition: to divide a solute between two immiscible solvents (with a boundary or interface)/AW; [1]

(b) (i) Mixture of iodine, ether and water is thoroughly shaken and left to equilibrate at constant temperature/AW; [1]

(25.00 cm³ of) each layer is titrated with a standard solution of sodium thiosulphate/AW; [1]

Partition coefficient is calculated using the formula

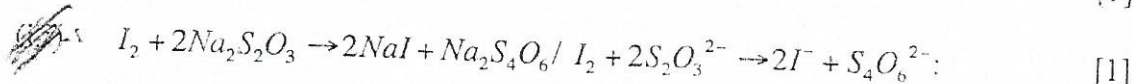
$$K = \frac{[I_2] \text{ in ether layer}}{[I_2] \text{ in aqueous layer}} / \text{AW}; \quad \text{ratio} \quad [1]$$

(f) (ii) $K = \frac{[I_2] \text{ in ether}}{[I_2] \text{ in water}}$

$$\therefore [I_2] \text{ in aqueous layer} = \frac{[I_2] \text{ in ether}}{K}$$

$$= \frac{0.0265}{42.7} \text{ mol dm}^{-3}$$

$$= 6.21 \times 10^{-4} \text{ mol dm}^{-3}; \quad [1]$$



$$\text{Moles } I_2 \text{ from ether layer} = \frac{1}{2} \text{ moles } S_2O_3^{2-}$$

$$= \frac{1}{2} \times \frac{26.50}{1000} \times 0.0500,$$

$$= 6.625 \times 10^{-4}; \quad [1]$$

$$\text{Concentration of } I_2 \text{ in ether} = \frac{6.625 \times 10^{-4}}{25.0 \times 10^{-3}}$$

$$= 0.0265 \text{ mol dm}^{-3}; \quad [1]$$

(iii) Moles I_2 in 25.0 cm^3 aqueous layer $= 6.21 \times 10^{-4} \times \frac{25}{1000}$
 $= 1.55 \times 10^{-5};$ [1]

Moles $S_2O_3^{2-}$ that reacted with $I_2 = 2 \times 1.55 \times 10^{-5} = 3.103 \times 10^{-5}$

$\therefore$ volume of $S_2O_3^{2-} = \frac{\text{moles}}{\text{conc}} = \frac{3.10 \times 10^{-5}}{1.00 \times 10^{-3}} = 0.031 \text{ dm}^3$
 $= 0.0331 \text{ dm}^3 / 31.03 \text{ cm}^3$ [1]

[Total: 10]

- 9 (a) At point (1) - solvent hexane evaporates; [1]
 - concentrating the solute; [1]

At point (2) Cold water cools the hexane vapour to condense falling into the inner tube; [1]

At point (3) hexane rises up through water because it is less dense than water; [1]

- as hexane rises up through water the solute partitions into it / solute is extracted; [1]

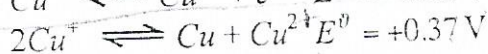
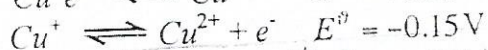
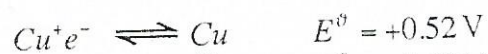
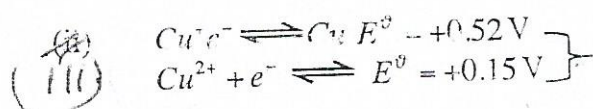
At point (4) the hexane layer overflows back into the heated flask with the solute; [1]
 [Any 6]

- (b) (i) - Steam is delivered into the reaction mixture to ---- heat up the mixture/evaporate the mixture/increase vapour pressure above the mixture; [1]
 - The internal pressure quickly builds up and equate to the external pressure boil at a temperature lower than usual; [1]
 - ~~X~~ Method is most suitable for samples that either have very high bpt/decompose near their bpt; [1]
 - mixture distils off together with water; [1]

4 max 3

- (ii) Water is separated by use of a separating funnel; [1]
 [Total: 10]

- 10 (c) (i) Simultaneous oxidation and reduction of a reactant; [1]



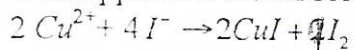
[1+1]



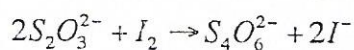
(11) Cu^{2+} more stable than Cu^+ in (aq):

cut less stable [1]

(b) White ppt in a brown solution [1]



white brown [1]



brown colour is discharged [1]

[Total: 10]

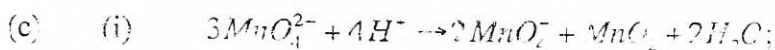
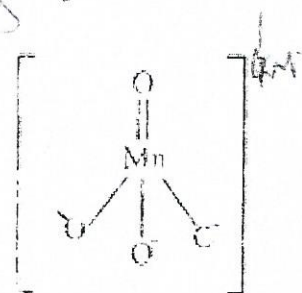
11 (a) MnO_4^{2-} ; green

MnO_4^- ; purple [1]

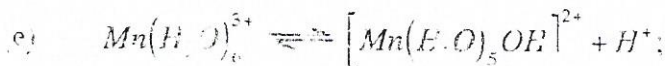
(b) (i) MnO_4^{2-} ; [1]

(ii) Tetrahedral; [1]

bond angle $109/109.5^\circ$; [1]



(ii) Making them strongly alkaline [1]



[Total: 6]

12 (a) (i) A $[Co(NH_3)_6]^{3+}$; [1]

B $[Co(NH_3)_6]^{3+}$; [1]

C $[CoCl_4]^{2-}$; [1]

D $[Co(CN)_6]^{4-}$; [1]

E $[Co(CN)_6]^{3-}$; [1]

(ii) - change in shape (octahedral to tetrahedral); [1]

- change in E^θ value (redox potential) [1]

		Co	N	H	Cl	
(b)	(i)					
	Mass in 100 g /g	25.2	24.1	5.1	45.6	
	Mole	$\frac{25.2}{58.9}$	$\frac{24.1}{14}$	$\frac{5.1}{1}$	$\frac{45.6}{35.5}$	
	=	0.428	1.714	5.1	1.29	
	Divide by smallest	1	4	12	3	[1]
	∴ Empirical formula is $CoN_4H_{12}Cl_3$;					[1]
(ii)	4 × NH_3 ligands and 3 × Cl^- ligands;					[1]
						[Total: 10]